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How Much Do Clinical and Microperimetric Findings Affect Reading Speed in Low Vision Patients with Age-related Macular Degeneration?

Deniz Altinbay [a,b], Aysun Idil [b], and Esra Sahli [b]

[a]Niv Eye Center, Department of Ophthalmology, Adana, Turkey; [b]Department of Ophthalmology, Vision Studies and Low Vision Rehabilitation Unit, Ankara University Faculty of Medicine, Ankara, Turkey

ABSTRACT

Purpose: To investigate the factors affecting the reading speed of patients with central scotoma due to age-related macular degeneration (AMD).

Materials and Methods: We included 63 eyes of 63 patients with AMD who applied to our low vision clinic between August 2018 and September 2019 in this prospective study. We evaluated socio-demographic characteristics, eye examination findings and Minnesota Low Vision Reading Test (MNREAD) results. We used the MAIA microperimeter device to evaluate the properties of the preferred retinal locus for fixation (PRL) of the patients. Evaluations included the assessment of the effects of all parameters on reading speed.

Results: The PRL was most commonly in the nasal (31%) and superior (26%) quadrants. Twenty-nine percent of the cases preferred the left visual field. PRL localization had no effect on reading speed, whereas, fixation stability, educational status, presence of foveal absolute scotoma, reading acuity and duration of reading interruption were found to have the most significant effects. Multiple regression analysis showed that reading speed decreased by 67 units in the presence of unstable fixation, by 17 units in the presence of foveal absolute scotoma, by 3 units with every 0.1 increase in logMAR value, and by 1.7 units with every 1-year increase in reading interruption. Additionally, being a university graduate was associated with an increased reading speed (by 18 units)

Conclusion: Increased reading performance is one of the factors that can improve quality of life. The factors found to affect the reading speed in the current study may guide the rehabilitation process in low vision patients.

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Low vision rehabilitation; microperimetry; reading speed; preferred retinal locus; central scotoma

Introduction

Decreased reading ability is the most common reason leading to patients' application to low vision centers.[1,2] Reading is a complex activity that is affected by many factors, including oculomotor control,[3,4] decreased fixation stability,[5–8] narrowed visual field,[9–13] comprehension speed,[14] near visual acuity,[3] contrast sensitivity,[3,7,15] print size[3] and age.[16] Therefore, the reading performance of a low vision patient is not always sufficient, even when the patient has good near visual acuity.[9,17,18]

In patients with age-related macular degeneration (AMD), the real fovea cannot perform its normal function; thus causing a shift of vision to a healthier area of the retina, defined as the preferred retinal locus (PRL).[10,19–22]

The microperimetry device has been used to evaluate the functional state of the macula and the fixation characteristics in these patients.[23,24] Fixation stability is an important factor that affects reading performance greatly, especially in the presence of central scotoma. This has been proven by various studies that have demonstrated a strong positive correlation between reading speed and fixation stability.[5–8] Although many studies have explored fixation stability and its effect on reading speed, there are only a limited number of studies investigating the effect of PRL location, distance to fovea and its relationship with scotoma and visual field on reading speed. Information concerning the location and stability of PRL is critical for the evaluation of the visual function, follow-up and treatment of patients with macular dysfunction.[24,25]

As mentioned before, there are a number of unrelated factors that contribute to a person's reading speed. Although some studies have explored the influence of various parameters on reading speed there are very few studies that have assessed the actual effects of these factors in a systematic manner that would enable direct comparisons with other studies and also between the parameters themselves. Therefore, in this study, we aimed to investigate the impact of microperimetric parameters, and sociodemographic and clinical characteristics on reading performance in patients with central scotoma due to AMD, and to determine the relationships between these parameters and reading speed. We obtained and reviewed these results with the intention that they could guide healthcare professionals performing near-sight rehabilitation.

Materials and methods

The ethical committee of Ankara University approved this prospective study. All steps and procedures were in accordance with the principles of the Helsinki Declaration. All individuals that participated in the study provided informed consent.

CONTACT Deniz Altinbay denizaltinbay01@gmail.com Ozel Niv Goz Merkezi, Sumer, 69023. Sk. No: 2/A, Seyhan, Adana 01140.
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We enrolled a total of 63 eyes diagnosed with AMD-related central vision loss from a group of 63 patients who had applied for visual rehabilitation to the Low Vision Rehabilitation Center of Ankara University Faculty of Medicine, between August 2018 and September 2019. The diagnoses of patients were either atrophic AMD or stable exudative AMD. We included patients who were older than 55 years, were at least primary school graduates, spoke Turkish as their native language, and did not have neurological diseases or mental disabilities that prevented them from reading or performing the MNREAD test. Additionally, only those with best-corrected visual acuity (BCVA) of between logMAR 1.3 and logMAR 0.5 were included in the patient group. Since visual acuity below 0.5 logMAR did not significantly affect reading performance, patients with BCVA better than 0.5 logMAR were not included in the study. Finally, patients who were illiterate, those that had previously undergone visual rehabilitation, and patients who used low-vision aids and/or had trained retinal locus (TRL) were excluded from the study.

We conducted detailed examinations in each patient, including BCVA measurement, biomicroscopic and fundus examination, low vision examination, contrast sensitivity test, lesion size, and quality of life scoring. Additionally, we asked patients about their reading habits prior to the disease and determined the time at which they had significantly lost their reading habit (being unable to continue their usual reading habits, or the inability to read books or newspapers). We recorded the time that had elapsed since the loss of reading habit (in years) and this duration was defined as 'reading interruption'. This data was obtained from patients and their relatives.

The BCVA was measured according to the Early Treatment Diabetic Retinopathy Study (ETDRS) chart and recorded as logMAR. Reading acuity was measured with the MNREAD test and recorded with 'M' value. The CSV-1000 test was used to test contrast sensitivity. The CSV-1000 test is traditionally applied with spatial fluorescence varying from 3, 6, 12 and 18 cpd (cycles per degree) values and grating from 2.5 meters.[26] However, since all subjects were patients with severe low-vision, we performed the test by halving the standard test distance and cpd values. We used the "Fundus autofluorescence (FAF)" method to calculate lesion size via the "Image Finder" program and recorded results in mm^2. The quality of life scale applied to patients was the Turkish-validated version of the NEI-VFQ-25.[27] The same technician (B.S.) conducted all evaluations with this questionnaire. From this scale, we recorded and evaluated the general score and the near-vision activities subscale score.

The Turkish version of the MNREAD reading test (MNREAD-TR) was the test used to determine reading performance. Idil et al. carried out the Turkish language reliability and validity study of this test.[28] This test is comprised of cards with print sizes representing logMAR 1.3 (8 M) to logMAR (−0.5) (0.12 M) (1 M = logMAR 0.4). The smallest print size that a person could read without making a significant mistake was used to calculate reading acuity (RA). Critical print size (CPS) was the smallest print size that could be read with maximum reading speed. Maximum reading speed (MRS) was the reading speed of a patient when unrestricted by print size. We used word per minute (wpm) values to depict the results of reading speed tests. The reading accessibility index (ACC) is calculated by dividing the reading speed obtained in the first 10 sentences of the MNREAD card by 200. These sentences represent the most frequently used print sizes in daily life. Since normal reading speed corresponds to 200 wpm, the normal ACC value is "1". ACC is a good indicator of reading speed in central scotoma.[5]

We used the MAIA Microperimetry device to evaluate fixation. P1 and P2 values showing the percentage of fixation points in a circle with a radius of 1° and 2° were used to analyze fixation stability. Definitions were as follows: 'stable fixation' when P1 was higher than 75%, 'relative unstable fixation' when P1 < 75% and P2 > 75%, and 'unstable fixation' when P1 and P2 were both lower than 75%.[29] The bivariate contour ellipse area (BCEA) was the elliptical area associated with the major and minor axes covered by fixative eye movements. The retinal sensitivity results range between 0 and 36 dB.

Macular Integrity Index (MII) is the assessment of macular integrity based on comparisons with age-matched responses. Categorization of MII values were as follows: a value under 40% was normal, 40–60% was suspicious and over 60% was abnormal.

We recorded patients' retinal sensitivity, MII, P1, P2, fixation stability, BCEA 63 and BCEA 95 values automatically, as they are part of the routine microperimetry analysis (Figure 1).

We performed some specific measurements for the study, in addition to the routine data obtained from the microperimetry analysis. We printed the grid-shaped screenshot showing the localization of the PRL on A4 paper and determined the placement of the PRL and calculated scotoma dimensions and the distance of the PRL to the fovea and the edge of the lesion. A single ophthalmologist (D.A.) performed all calculations in order to prevent inter-operator variance. The operator measured the distance between the PRL and Fovea in millimeters and calculated degree-equivalents. In cases with uncertain fovea location, we estimated and marked the fovea location by calculating from the middle of the optic disc to 15.5° horizontally (towards the temporal) and 1.3° towards the inferior.[30] We classified the location of the PRL as superior, inferior, temporal, nasal or central (central 4°), with respect to the center of the fovea[31] (Figure 2).

We used the IBM SPSS Statistics Version 20.0 package program for all statistical analyses. We defined the threshold for statistical significance as $p < .05$ in all tests. The Shapiro Wilk test was used to test the normality of the distribution of numerical measurements. The independent samples t-test and the Mann–Whitney U test were used in the comparison of numerical measurements among two groups, with respect to the normality of distribution. We used the One-Way Variance Analysis or the Kruskal Wallis test for general comparisons of numerical measurements in more than two groups – again according to the normality of distribution. The Bonferroni, Scheffe or Tamhane methods were used for post-hoc corrections after identification of significant differences in >2 group comparisons – depending on variance homogeneity.

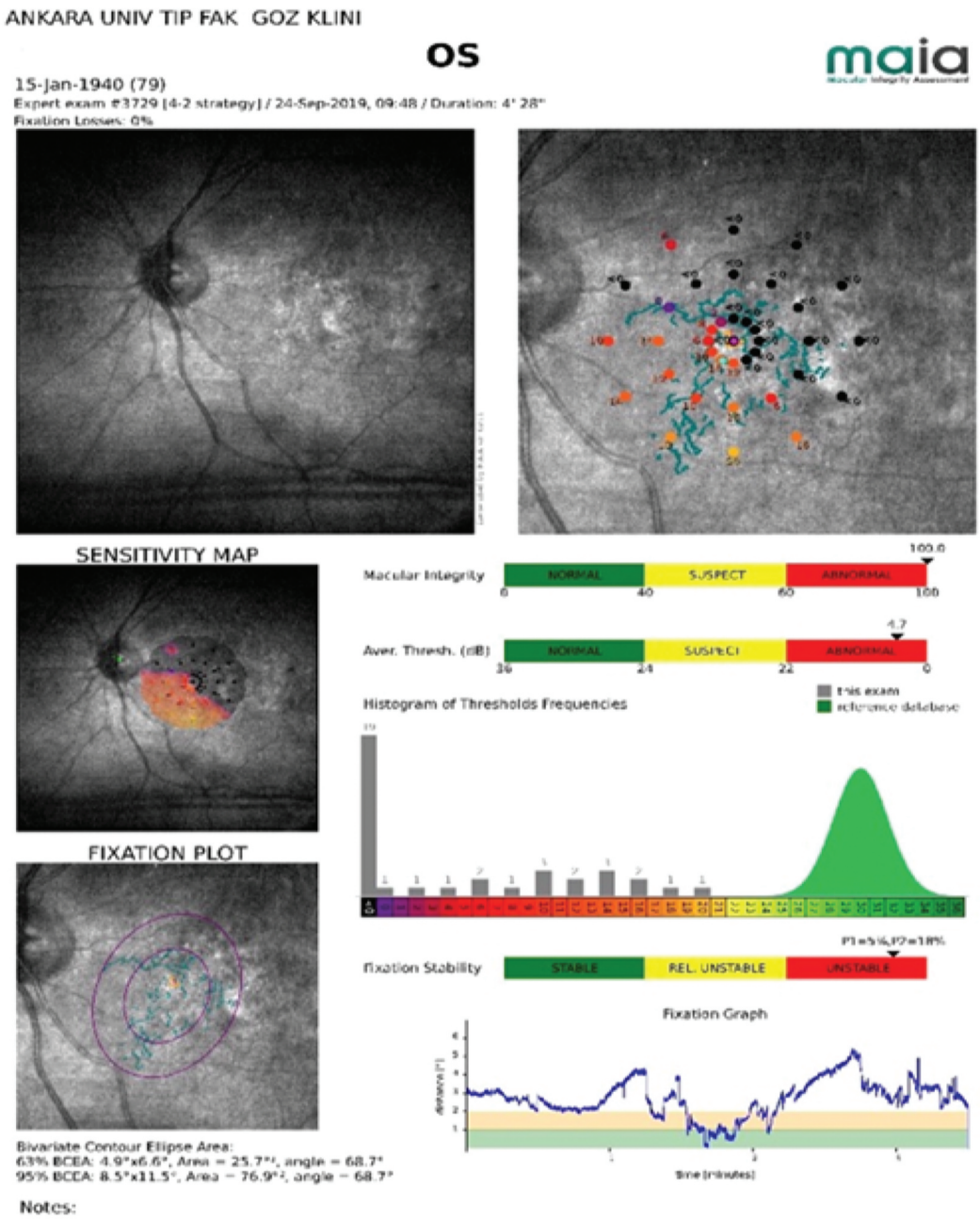


Figure 1. Example of microperimetry output in a case with AMD.

Results

The study included a total of 63 patients with advanced AMD and low vision, aged between 58 and 93 years. The eye with better visual acuity was the left eye in 36 patients and the right eye in 27 patients. The percentage of patients with atrophic AMD (48%) and exudative AMD (52%) were similar. Patients with undergraduate education had higher MRS values (p = .008). Additionally, MRS values were positively correlated with total score (p = .050) and near-vision activities subscale score (p = .027) of the quality-of-life questionnaire, and negatively correlated with duration of reading interruption (p = .014). The sociodemographic characteristics of the cases are shown in Table 1.

The mean distant BCVA value was 0.66 ± 0.2 logMAR, while the mean near BCVA was 2.63 ± 2.46 M. The mean lesion size was 7.162 ± 4.671 mm2 (min–max: 1.110–24.568 mm2). The mean contrast sensitivity results/values at low frequencies (1.5 cpd and 3 cpd) were 12.3 ± 11.2 dB and 21.0 ± 15.1 dB, respectively. For the medium frequencies (6 cpd and 9 cpd), the results were 10.62 ± 9.49 dB and 4.46 ± 3.93 dB, respectively.

When the correlations between eye examination findings and reading speed were assessed, we found negative correlations between MRS values and both the distant and near acuities (p = .002 and p = .003, respectively). MRS was found to be negatively correlated with the amount of high-powered

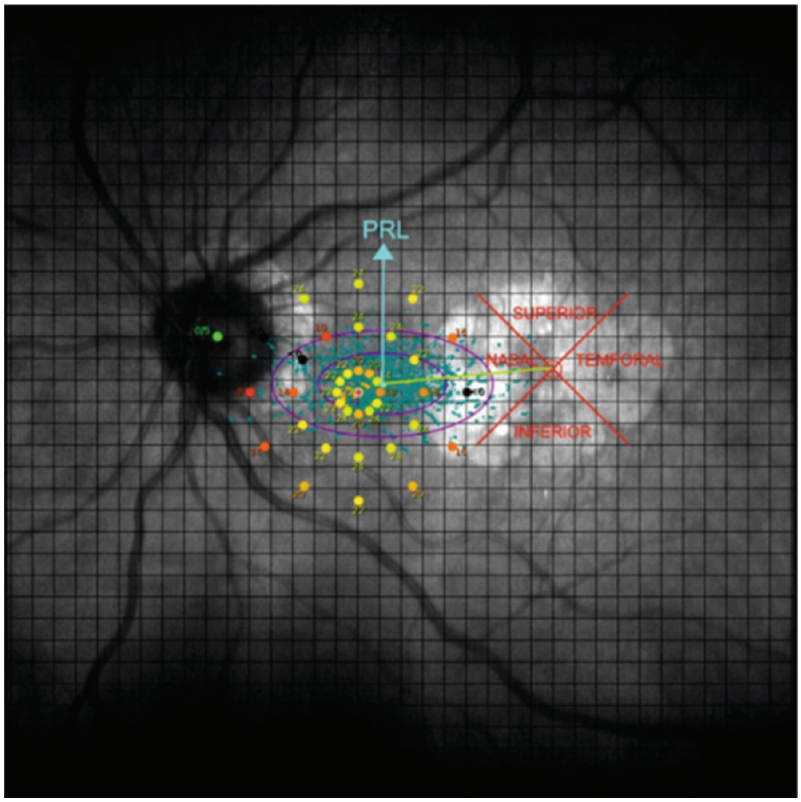


Figure 2. Fovea-centered macula divided into four quadrants, PRL localizations.

Table 1. Distribution of the sociodemographic characteristics of the cases.

Feature		Number (Percent)
Gender	Male	41 (65)
	Female	22 (35)
Educational Status	Primary Education	21 (33,3)
	High School Education	21 (33,3)
	Bachelor's Education	21 (33,3)
		Mean ± SD
Age		77,49 ± 8,56
Symptom Duration (years)		4,73 ± 4,09
Reading Habit (hour)		2,1 ± 2,23
Reading Interruption (years)		2,58 ± 3,08
Quality of Life Overall Score		47,5 ± 11,4
Near-vision Activities Subscale Score		29,0 ± 12,4

Table 3. Distribution and average results of the results obtained from microperimetry.

Feature		Mean ± SD
MII		99,8 ± 1,6
Average Retinal Sensitivity (dB)		10,8 ± 6,5
P1		25,5 ± 20,4
P2		56,2 ± 23,4
		Number (Percent)
Fixation Stability	Stable	3 (%5)
	Relatively Stable	17 (%27)
	Unstable	43 (%68)
		Mean ± SD
BCEA 63		17,3 ± 10,8
BCEA 95		51,6 ± 32,4

near addition (p = .020). It was also negatively correlated with contrast sensitivity at low spatial frequencies (p = .001).

We found that MRS ranged from 19 wpm to 160 wpm. MRS was lower than 80 wpm in 42 (66%) and lower than 40 wpm in 17 (27%) patients. The MNREAD-TR results of the cases included in the study are shown in Table 2. We found that the RA, CPS and ACC parameters were significantly correlated with MRS (p < .001, p = .001 and p < .001, respectively).

Table 3 depicts the results obtained with routine microperimetry analyses. According to these results, unstable and relative unstable fixation stability was found in 95% of our study group (n = 60). Only 3 patients had 'stable' fixation, while 17 had relative unstable fixation.

According to the measurements based on the microperimetry printout, we found that the mean PRL-fovea distance was 5.15° ± 3.31° (between 0.75° and 14.2°). When PRL localizations were assessed, we found the nasal quadrant was the most common (n = 19, 31%), followed by superior (n = 16, 26%), inferior (n = 10, 16%), central (n = 10, 16%) and temporal (n = 7, 11%). In terms of scotoma, 42 patients had absolute scotoma and 11 had relative scotoma. Absolute scotoma size ranged between 0° and 62°. Absolute scotoma size was greater than 22° in 16 (25%) patients. MRS was associated with PRL-fovea distance, PRL-lesion edge distance and scotoma size (p < .001, p = .046 and p = .001, respectively).

When the localization of PRL and scotoma were assessed, 18 (29%) cases had a PRL located on the left side of the scotoma, 14 (22%) cases had a PRL within the scotoma, 9 (14%) cases had a PRL located under the scotoma, 7 (11%)

Table 2. MNREAD reading card results of cases.

Feature	Mean ± SD (interval)
Reading Accuracy (RA) (logMAR)	0,86 ± 0,34 (0,12–1,80)
Critical Print Size (CPS) (logMAR)	1 ± 0,35 (0,32–1,90)
Maximum Reading Speed (MRS) (wpm)	70,2 ± 37,23 (19–160)
Reading Accessibility Index (ACC)	0,27 ± 0,21 (0–0,69)

Table 4. Factors that affect reading speed the most.

Feature		Maximum Reading speed Mean ± SD	p
Educational Status	Primary School graduate	61,2 ± 36,4	0,008
	High School graduate	58,9 ± 32,1	
	Bachelor's Degree	90,5 ± 35,9	
Fixation Stability	Stable	148,7 ± 12,7	0,003
	Relatively Stable	80,9 ± 41,1	
	Unstable	60,5 ± 28,5	
Foveal Absolute Scotoma	Yes	97,8 ± 30,9	<0,001
	No	56,4 ± 32,3	

Correlations with Maximum Reading Speed	Correlation Coefficient	p
Reading Interruption (years)	−0,307	0,014
Reading Acuity (logMAR)	−0,533	<0,001

cases had a PRL located on the right side of the scotoma, and 4 (%6) cases had a PRL located above the scotoma. Eleven (17%) cases had no absolute scotoma. In different comparisons with respective groups, it was found that reading speeds were lowest in patients with PRL in the center of the scotoma (p = .001), and also in patients who had foveal absolute scotoma (p < .001). Finally, we also found that 18 (29%) patients preferred the left visual field, while 8 (13%) patients preferred the right visual field.

Multiple regression analysis showed that reading speed decreased by 67 units in the presence of unstable fixation, by 17 units in the presence of foveal absolute scotoma, by 3 units with every 0.1 increase in logMAR value, and by 1.7 units with every 1-year increase in reading interruption. Additionally, being a university graduate was associated with an increased reading speed (by 18 units) (Table 4).

Discussion

We investigated factors that affect the reading performance of patients with AMD and determined that there were relationships between reading speed and parameters, such as fixation stability, presence of foveal absolute scotoma, reading acuity, education level, and duration of reading interruption. The factor that decreased reading speed the most was found to be the presence of unstable fixation. This result shows the importance of fixation stability in increasing reading speed in near visual rehabilitation, as it is apparent that near visual acuity is not sufficient for the evaluation of reading performance.

In addition to the parameters obtained from microperimetry analysis and the automated calculation results, we also evaluated other factors that may affect reading speed, such as sociodemographic characteristics, eye examination findings, contrast sensitivity and MNREAD parameters. We also conducted multiple linear regression analysis to identify factors that were independently associated with reading speed. To our knowledge, this study is the only study in the literature that focused on this rarely researched subject via the measurement of an extensive set of parameters with a study group that had a relatively high number of patients.

Previous studies have reported that reading speed should be at least 80 wpm for fluent reading and 40 wpm is only sufficient for spot reading.[11] Reading speed is significantly lower in patients with central scotoma compared to the normal population. Complete loss of reading ability is common in these patients. In our study, the MRS of the patients ranged between 19 and 160 wpm, while MRS was lower than 80 wpm in 42 (66%) and lower than 40 wpm in 17 (27%) patients.

The strong relationship between fixation stability and reading speed has been demonstrated in many studies.[5,7,8] Crossland et al. reported that fixation stability was responsible for around 54% of the change in reading speed in patients with AMD and juvenile macular degeneration. Besides this, they also reported that the decrease in reading speed was not associated with fixation instability alone.[6] In our study, in accordance with the literature, a statistically significant relationship was found between fixation stability and MRS, but unstable fixation was not found to be the only factor that reduced reading speed. Having foveal absolute scotoma and duration of reading interruption also influenced reading speed. The reading speed was also affected by the education level of the patients.

In the presence of central scotoma, visual span narrows; therefore, patients often rely on guessing words in order to be able to read,[13,14] which increases the importance of visual abilities. We evaluated the influence of reading interruption on the speed of reading because we considered this factor as a confounding variable that would alter reading speed due to its possible adverse effects on cognitive functions. According to our analysis, each year of reading interruption decreased reading speed by 1.7 units. If there are periods when the patient does not read, it is very likely that his/her visual skills, such as visual memory and visual completion ability, will deteriorate."

Various studies have shown that scotoma size affects reading speed.[11,13,14] Scotoma size is especially important in terms of identifying and perceiving a word that is being read. In central scotoma, individuals can perform normal reading in the absence of foveal absolute scotoma with the presence of 2°–5° of the field of view in the horizontal plane (2° to the right and left of fixation).[32] In their study, Whittaker et al.[11] have reported that field of view (number of letters visible) and scotoma size were important for reading, and that fluent reading was not possible in the presence of scotoma size greater than 22°. Ergun et al.[13] found a negative correlation between absolute scotoma size and reading speed (r = −0.48, p = .023). We also found a negative correlation between absolute scotoma size and MRS (r = −0.422, p = .001). Various factors, such as scotoma location, fixation stability and other parameters, could explain the relatively lower correlation between these parameters (relative to other correlations) in the present study. Absolute scotoma size ranged between 0° and 62° in our study, and it was greater than 22° in 16 cases (25%). When we evaluated the reading speed of these cases, we found that MRS values were lower than 80 wpm in 14 (87.5%) of these individuals, indicating that they could not read fluently.

The presence of foveal absolute scotoma also decreases reading performance.[32] A statistically significant relationship was found between reading speed and foveal absolute scotoma in this study. The multiple regression analysis revealed that the

presence of foveal scotoma caused a decrease of 17 units in MRS value. This is largely due to the fact that the fovea plays a critical role in visual acuity, as it is the area with the highest retinal resolution.[33]

It has been reported that 77–100% of low-vision patients with poor central vision experience adaptations through various mechanisms. In most cases, this adaptation is the development of a PRL.[10,24,34,35] In the current study, 98% of the cases had developed a PRL; 84% of these were extrafoveal and 16% were foveal. However, foveal fixation is not sufficient for good reading speed. When there is retinal sensitivity loss in the area surrounding the fovea, visual acuity is significantly reduced, even when fixation is central.[36] The significant relationship between retinal sensitivity loss and reading speed has been shown in our study.

There are studies reporting that the stability of PRL decreases as it moves away from the fovea.[24,31,37] PRL-fovea distance has been reported to vary between 1.0° and 9.0°. In this study, the PRL-fovea distance was found to range between 0.75° and 14.2°, and we also found that both fixation stability and reading speed reduced with increased distance between the PRL and fovea. This is due to decreased retinal resolution and sensitivity in the PRL region.[33]

The majority of studies evaluating patients with AMD report that PRL is most frequently localized in the nasal and superior quadrants.[20,31] For instance, Erbezci and Öztürk found PRL most frequently in the nasal quadrant (29.2%) and secondly in the superior quadrant.[31] Very similarly, we found that PRL was most commonly present in the nasal quadrant (31%) and the superior quadrant (26%) in our group of patients. Additionally, studies show that patients with central scotoma most commonly prefer the left visual field,[10,20,31,34,38] followed by the inferior visual field.[20,31,34] Our results were also in agreement with the majority of published literature.

There are several studies reporting an association between reading speed and the localization of PRL.[34,39] For instance, there are studies suggesting that PRL localization in the superior quadrant increases reading speed.[20,38] However, others suggest a lack of meaningful relationship.[35,38,40] Conforming to the latter group of studies, we also found relationship between reading speed and PRL location. In agreement with our results, Fletcher et al. also did not find a relationship between reading speed and the localization of PRL. They concluded that, apart from the patient's cognitive capabilities, the fixation and saccadic movements of the eyes were the primary parameters that were important for reading in patients with central scotoma.[38]

Restoring reading ability in people with low vision due to central scotoma is one of the main goals of low-vision rehabilitation. Thus, in these cases, it is important to identify correctable or modifiable factors that influence reading speed the most. The present study has found fixation stability to be a modifiable factor that affects reading speed the most. Studies using microperimetric biofeedback signals have been shown to increase fixation stability and reading speed.[41,42] It was reported by Tarita-Nistor et al. that the most appropriate PRL should be created in each patient, since their findings demonstrated a speed increase of 38%, as well as increases in reading acuity and CPS.[41]

The strength of this study is the analysis of a great number of variables that may have been associated with or could have influenced reading speed, including scotoma size, contrast test and duration of reading interruption.

The number of participants included in the study and the lack of contrast sensitivity measurement via a test standardized for individuals with low-vision are limitations of the study. If there indeed is a PRL location that influences reading speed, we believe that larger studies which include additional analyses for reading direction (left-to-right, top-to-bottom) may be valuable to elucidate the role of PRL localization on reading speed.

Conclusion

The reading speed of low-vision individuals with AMD is often insufficient for fluent reading, and the recuperation of reading speed is an important aspect of low-vision rehabilitation. Since reading performance appears to be significantly affected by fixation stability in patients with AMD, near vision rehabilitation therapies may benefit from the incorporation of methods such as microperimetric biofeedback training to increase fixation stability.

Highlights

- Reading speed is strongly associated with fixation stability.
- Reading speed is associated with a number of factors at varying degrees.
- The presence for absolute foveal scotoma is more determinant in the reading speed then the size of the scotoma.
- There is no relationship between reading speed and the retinal locus preferred for fixation.

Author contributions

Conceptualization; Deniz Altinbay, Aysun Idil, **Data Curation**; Deniz Altinbay, Esra Sahli, **Formal Analysis**; Deniz Altinbay, Esra Sahli, Aysun Idil, **Investigation;** Deniz Altinbay, Esra Sahli, **Methodology;** Deniz Altinbay, Esra Sahli, Aysun Idil **Project administration**; Aysun Idil, **Supervision;** Aysun Idil, **Validation;** Aysun Idil, **Visualization;** Esra Sahli, **Roles/Writing-original draft**; Deniz Altinbay, **Writing-review&editing**; Deniz Altinbay, Esra Sahli, Aysun Idil

Disclosure statement

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ORCID

Deniz Altinbay http://orcid.org/0000-0002-3976-4361
Aysun Idil http://orcid.org/0000-0002-5979-9158
Esra Sahli http://orcid.org/0000-0002-1355-7284

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